FINITE PRESENTATION OF A MODULE IS EQUIVALENT UPSTAIRS, ON THE SUPPORT AND ON THE FINITE COVER, IN THE ELEMENTARY DEVISSAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; the counted span uses only displayed formulas, but the chapter preamble is retained unedited rather than trimmed. No macro was invented or redefined.

Editorial note. The same-chapter referenced result is located at its \label line inside the pinned chapter file; the nine cross-chapter statements are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The elementary devissage diagram and its notation. Same chapter, Lemma labelled lemma-elementary-devissage, source line 494 (Stacks tag 057S); the statement of this card literally begins with 'Assumptions and notation as in' that lemma. Supplies the objects x, x', z', y', the morphisms g, i, pi and the modules G, F appearing throughout; the equivalences themselves are the counted content.
- Permanence of finite presentation under composition, pullback and finite extension of local rings. Morphisms, Lemmas morphisms-lemma-composition-finite-presentation (01TR) and -finite-presentation-permanence (02FV); Modules, Lemma modules-lemma-pullback-finite-presentation (01BQ); Algebra, Lemmas algebra-lemma-descend-properties-modules (03C4) and algebra-lemma-finite-finitely-presented-extension (0564). Cited at proof lines 627-684 as the input of each equivalence; the organisation of the two chains of implications and the descent along the etale covering stays in the counted span.
- Descent of finite presentation along an etale covering and along the finite morphism with a single-point fibre. Descent, Lemmas descent-lemma-finite-presentation-descends (05B0) and descent-lemma-finite-finitely-presented-module (05B4); More on Morphisms, Lemma more-morphisms-lemma-finite-morphism-single-point-in-fibre (05B8); Morphisms, Lemma morphisms-lemma-etale-open (03WT). Supply openness of g, descent of finite presentation through the etale covering and the local-ring form of the finite morphism with one point in the fibre; none of them contains the equivalence being proved.

Source reference. Original lemma 057S: sources/stacks-flat/pinned-original-source.tex, source lines 494–494. Same-chapter pointer named in the first line of the counted

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statement (line 600) as the source of the assumptions and notation; the equivalence proof itself is retained verbatim in the delivered body and is the counted content of this card.

Source reference. Original lemma 01TR: sources/stacks-chow/tags.txt, source lines 2384–2384. Cross-chapter standard prerequisite invoked once at proof line 627 to see that Z' -> S is locally of finite presentation as a composite.

Source reference. Original lemma 02FV: sources/stacks-chow/tags.txt, source lines 3206–3206. Cross-chapter standard prerequisite invoked once at proof line 630 to transfer local finite presentation from Z' -> S and Y' -> S to pi.

Source reference. Original lemma 01BQ: sources/stacks-chow/tags.txt, source lines 1735–1735. Cross-chapter standard prerequisite invoked once at proof lines 639-640 for the implication (1) => (4).

Source reference. Original lemma 03WT: sources/stacks-chow/tags.txt, source lines 5124–5124. Cross-chapter standard prerequisite invoked once at proof lines 642-643 to know that g(U') is an open neighbourhood of x, which is what makes U' -> g(U') an etale covering.

Source reference. Original lemma 05B0: sources/stacks-chow/tags.txt, source lines 6868–6868. Cross-chapter standard prerequisite invoked once at proof line 647 for the implication (4) => (1) by descent along the etale covering.

Source reference. Original lemma 05B4: sources/stacks-chow/tags.txt, source lines 6872–6872. Cross-chapter standard prerequisite invoked once at proof line 649 for the equivalences between (4), (2) and (3).

Source reference. Original lemma 05B8: sources/stacks-chow/tags.txt, source lines 6876–6876. Cross-chapter standard prerequisite invoked once at proof lines 651-652 (and again in the local-ring chain at 679-680) for the behaviour of the finite morphism pi with a single point in the fibre.

Source reference. Original lemma 03C4: sources/stacks-chow/tags.txt, source lines 4371–4371. Cross-chapter standard prerequisite invoked once at proof line 668 for descent of finite presentation along the faithfully flat local ring map O_{X,x} -> O_{X',x'}.

Source reference. Original lemma 0564: sources/stacks-chow/tags.txt, source lines 6697–6697. Cross-chapter standard prerequisite invoked twice, at proof lines 674 and 684, for the criterion for finite presentation along a finite ring extension; the two local-ring comparisons it supports are retained in the counted span.

Lemma 0.1. *Assumptions and notation as in Lemma 057S. If f is locally of finite presentation then π is of finite presentation. In this case the following are equivalent*

(1) *$\mathcal{F}$ is an $\mathcal{O}_X$-module of finite presentation in a neighbourhood of x,*
(2) *$\mathcal{G}$ is an $\mathcal{O}_{Z'}$-module of finite presentation in a neighbourhood of z', and*
(3) *$\pi_*\mathcal{G}$ is an $\mathcal{O}_{Y'}$-module of finite presentation in a neighbourhood of y'.*

Still assuming f locally of finite presentation the following are equivalent to each other

(a) *$\mathcal{F}_x$ is an $\mathcal{O}_{X,x}$-module of finite presentation,*
(b) *$\mathcal{G}_{z'}$ is an $\mathcal{O}_{Z',z'}$-module of finite presentation, and*
(c) *$(\pi_*\mathcal{G})_{y'}$ is an $\mathcal{O}_{Y',y'}$-module of finite presentation.*

Proof. Assume f locally of finite presentation. Then $Z' \to S$ is locally of finite presentation as a composition of such, see Morphisms, Lemma 01TR. Note that $Y' \to S$ is also locally of finite presentation as a composition of a smooth and an étale morphism. Hence Morphisms, Lemma 02FV implies π is locally of finite presentation. Since π is finite we conclude that it is also separated and quasi-compact, hence π is actually of finite presentation.

To prove the equivalence of (1), (2), and (3) we also consider: (4) $g^*\mathcal{F}$ is a $\mathcal{O}_{X'}$-module of finite presentation in a neighbourhood of x'. The pullback of a module of finite presentation is of finite presentation, see Modules, Lemma 01BQ. Hence (1) $\Rightarrow$ (4). The étale morphism g is open, see Morphisms, Lemma 03WT. Hence for any open neighbourhood $U' \subset X'$ of x', the image $g(U')$ is an open neighbourhood of x and the map $\{U' \to g(U')\}$ is an étale covering. Thus (4) $\Rightarrow$ (1) by Descent, Lemma 05B0. Using Descent, Lemma 05B4 and some easy topological arguments (see More on Morphisms, Lemma 05B8) we see that (4) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3).

To prove the equivalence of (a), (b), (c) consider the ring maps

$$\mathcal{O}_{X,x} \to \mathcal{O}_{X',x'} \to \mathcal{O}_{Z',z'} \leftarrow \mathcal{O}_{Y',y'}$$

The first ring map is faithfully flat. Hence $\mathcal{F}_x$ is of finite presentation over $\mathcal{O}_{X,x}$ if and only if $g^*\mathcal{F}_{x'}$ is of finite presentation over $\mathcal{O}_{X',x'}$, see Algebra, Lemma 03C4. The second ring map is surjective (hence finite) and finitely presented by assumption, hence $g^*\mathcal{F}_{x'}$ is of finite presentation over $\mathcal{O}_{X',x'}$ if and only if $\mathcal{G}_{z'}$ is of finite presentation over $\mathcal{O}_{Z',z'}$, see Algebra, Lemma 0564. Because π is finite, of finite presentation, and $\pi^{-1}(\{y'\}) = \{x'\}$ the ring homomorphism $\mathcal{O}_{Y',y'} \leftarrow \mathcal{O}_{Z',z'}$ is finite and of finite presentation, see More on Morphisms, Lemma 05B8. Hence $\mathcal{G}_{z'}$ is of finite presentation over $\mathcal{O}_{Z',z'}$ if and only if $\pi_*\mathcal{G}_{y'}$ is of finite presentation over $\mathcal{O}_{Y',y'}$, see Algebra, Lemma 0564. $\square$